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Reflection Document

1. Where did your agent fail in its first attempt?

In the first attempt, my agent had several limitations. The AI model was responsible for generating both slide titles and content, which often resulted in generic or inaccurate information instead of reliable, fact-based content. The web search functionality was also not fully dynamic and depended on limited or predefined responses, making it ineffective for new topics. Additionally, the system followed a fixed sequence of steps, behaving more like a hardcoded script rather than an intelligent agent. There were also formatting issues in the generated PowerPoint, such as overcrowded slides and inconsistent styling.

✅ Resolution:

To address these issues, I redesigned the system architecture. I restricted the AI model's role to only generating slide titles, ensuring it focuses on planning rather than content creation. For content generation, I integrated a real-time web search MCP tool so that all bullet points are based on actual data instead of generated guesses.

I also implemented a proper agentic loop where, for each slide, the system fetches relevant data, processes it, and builds the slide step by step. To resolve formatting problems, I enforced limits on the number of bullet points and improved layout handling. Additionally, I introduced a theme system to ensure consistent design across all slides. These changes made the system more accurate, scalable, and reliable.

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2. How did MCP prevent you from writing hardcoded scripts?

Initially, the system logic was becoming hardcoded, meaning the sequence of operations was fixed and not flexible. This made it difficult to handle different topics or extend the system with new features. The system lacked modularity and behaved more like a predefined script rather than a dynamic AI agent.

✅ Resolution:

The introduction of the Model Context Protocol (MCP) resolved this issue by enabling a modular and tool-based architecture. Instead of embedding all logic directly into the code, I defined separate tools such as web search and PowerPoint generation.

The agent now dynamically interacts with these tools, deciding when to fetch information and when to build slides. This eliminated the need for hardcoded flows and allowed the system to handle any input topic effectively. MCP also improved scalability and maintainability, as each tool can be updated independently. Overall, MCP transformed the system into a true autonomous agent capable of flexible and dynamic decision-making.

Issues Faced and Resolutions

Issue 1: Generic or Fallback Content Generation

In the initial implementation, the slide content often included generic or fallback text instead of meaningful, real-world information. This reduced the quality and reliability of the generated presentations.

Resolution:
I enforced the use of the MCP search_web tool for every slide, ensuring that all content is fetched from real-time web data. I also removed fallback-based content generation so that only relevant and factual bullet points are included.

Issue 2: Model Output Noise and Irrelevant Text

The model initially generated noisy outputs such as incomplete sentences, random phrases, or unwanted text like sources (e.g., “WSJ”, URLs), which made the slide content unprofessional.

Resolution:
I improved prompt engineering by providing strict instructions to the model (e.g., fixed number of bullet points, no extra text, no sources). Additionally, I implemented an output cleaning mechanism to filter out irrelevant or malformed lines before passing the content to the MCP tools.

Issue 3: Slide Content Overflow and Poor Layout

The generated slides sometimes contained too much text, causing overflow and making the presentation difficult to read.

Resolution:
I introduced layout constraints such as limiting the number of bullet points per slide and controlling the length of each point. This ensured that the content fits properly within the slide and maintains readability.

Issue 4: Inconsistent Theme and Styling

Initially, the presentation lacked consistent styling, with themes not being applied uniformly across all slides.

Resolution:
I implemented a centralized theme configuration system and ensured that the selected theme is applied globally before creating the presentation. This guaranteed consistent colors, fonts, and layout across all slides.